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Effects of parasite specificity and previous infestation of hosts on the feeding and reproductive success of rodent-infesting fleas

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Summary

1. We examined whether identity of the rodent host and previous infestation of the host affect feeding and reproduction of fleas. We predicted that feeding and reproductive success of fleas would be higher when feeding on (i) a typical host than on an atypical host; and (ii) a pristine host than on a host previously exposed to parasitism. We also predicted that the negative effect of previous infestation would not be manifested in a host-specific flea feeding on its typical host.

2. To test these predictions, we measured blood meal size, egg production and latency of oviposition in host-specific *Parapulex chephrenis* and host-opportunistic *Xenopsylla ramesis* during seven daily feedings on the Egyptian spiny mouse *Acomys cahirinus* (typical host of the former) and Wagner's gerbil *Dipodillus dasyurus* (typical host of the latter).

3. Blood meal size in *P. chephrenis* did not depend on either host species or previous host infestation with fleas. However, when this flea fed on *D. dasyurus* as opposed to *A. cahirinus*, blood meal size increased to the end of 7-day period. *Xenopsylla ramesis* took larger blood meals (i) during the first feeding; (ii) from *D. dasyurus* than from *A. cahirinus*; and (iii) from pristine than from previously infested *D. dasyurus*, but the blood meals taken from pristine and previously infested *A. cahirinus* were similar.

4. Egg production of *P. chephrenis* was significantly higher and oviposition started earlier when it fed on *A. cahirinus*; this was true for *X. ramesis* when it fed on *D. dasyurus*. Surprisingly, *P. chephrenis* laid more eggs and started oviposition earlier when it fed on previously infested rodents. However, egg production in *X. ramesis* and start of oviposition were similar in pristine and previously infested hosts.

5. These results suggest that the response of a parasite to acquired immunity of a host may depend on the host species, level of parasite host specificity as well as the degree of 'tightness' of a particular parasite–host association.

Key-words: blood meal size, egg production, flea, resistance, rodent host

Introduction

A host defends itself against parasites mainly by its immune system. Immune defence mechanisms are either innate or acquired. Generally, innate immunity is non-specific (e.g. non-specific phagocytosis of parasites by polymorphonuclear granulocytes), whereas acquired immunity is an induced response characterized by specificity for the sensitizing immunogen (e.g. procession of antigens by Langerhans cells and present them to T cells which initiate inflammatory and antibody responses) (see Wakelin 1996; Wikel 1996 for reviews).

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Acquired immunity is believed to play a major role in developing resistance of the host against parasites.

The host immune responses to saliva of the haematophagous arthropods can neutralize salivary molecules essential for successful feeding (e.g. Rechav & Dauth 1987). Haematophagous arthropods, in turn, have developed countermeasures to harmful immune responses of their hosts (Wikel & Alacron-Chaidez 2001). For example, saliva of the tick *Amblyomma americanum* contains a homologue of a human cytokine, which may counteract the host's immune responses (Jaworski *et al.* 2001). Attempts by a host to defend itself against parasites and of a parasite to neutralize the host's defence represent, thus, a classical example of evolutionary and ecological arms race between a victim (host) and an enemy (parasite).

Host-focused approaches prevail in most studies of resistance against parasites, including haematophagous arthropods. In other words, the defence success of a host is measured via various parameters of the host itself. However, it is the parasite's response to the host defence that is important from both evolutionary and ecological perspective of host–parasite relationships. Consequently, the studies of the defence success of a host should be parasite-focused, that is, it should be measured in terms of loss of fitness of the parasite.

An individual host attacked repeatedly by an ectoparasite often develops acquired resistance against this ectoparasite (Willadsen 1980; Rechav 1992), which is manifested by decreased feeding and reproduction of the ectoparasite (e.g. Fielden, Rechav & Bryson 1992). However, a host–parasite association is often a product of a long history of adaptation of a parasite to a host or group of hosts (Combes 2001). As a result, many parasites are highly host-specific (Poulin 1992; Combes 2001). One of the adaptations of a parasite to a given host can be that a specific parasite does not stimulate the host's immune response and resistance acquisition (Ribeiro 1989; Beaty & Marquardt 1996). Although among-host differences in ability to acquire resistance against a parasite have been reported (Trager 1939; Randolph 1979), the relationship between acquired resistance of a host and the degree of specificity of a parasite to this host is poorly understood. In addition, the evidence of acquired resistance against parasites other than ticks is indirect and based on responses by hosts rather than by parasites (Khokhlova *et al*. 2004).

Here we asked whether the host species identity and acquired resistance affect feeding and reproductive characteristics of host-specific and host-opportunistic parasites. To answer this question we studied feeding and reproductive variables in two flea species, host-specific *Parapulex chephrenis* and host-opportunistic *Xenopsylla ramesis*, when feeding on two rodent species, the Egyptian spiny mouse *Acomys cahirinus* and Wagner's gerbil *Dipodillus dasyurus*. *Acomys cahirinus* is a typical host for *P. chephrenis*; this flea very rarely parasitizes any other rodent, including congeneric and co-habitating *A. russatus* (Krasnov *et al*. 1997, 2005a). *Xenopsylla ramesis* is a host generalist exploiting a variety of rodent species. In the Negev Desert, it was routinely collected from eight host species (B. R. Krasnov & G. I. Shenbrot, unpublished data). Main hosts of this flea belong to the subfamily Gerbillinae, although it has been recorded occasionally on spiny mice. We measured feeding efficiency via blood meal size and reproductive success via egg production and latency of oviposition, in fleas after they fed on rodents that either had never been parasitized by any flea species or were repeatedly exposed to flea parasitism prior to measurements. Exposure of rodents to flea parasitism was assumed to induce acquired resistance against fleas (Fielden *et al*. 1992; Khokhlova *et al*. 2004).

We predicted that feeding efficiency and reproductive success of fleas will be higher when a flea feeds on (i) a typical host; and (ii) hosts that had not been exposed to fleas. We also predicted that rodents will acquire resistance against a non-specific parasite, but will not acquire resistance or acquire only weak resistance against a specific parasite. Consequently, we expected that the negative effect of previous exposure to parasitism will be manifested in *X. ramesis* feeding on either host and in *P. chephrenis* feeding on a gerbil, but not on a spiny mouse.

Methods

MAINTENANCE OF FLEAS AND RODENTS

We obtained fleas from our laboratory colonies started in 1999 from field-collected specimens on *D. dasyurus* and *Meriones crassus* (*X. ramesis*) and *A. cahirinus* (*P. chephrenis*). Fleas were reared on their characteristic rodent hosts, *X. ramesis* on *D. dasyurus*, and *M. crassus* and *P. chephrenis* on *A. cahirinus*, using procedures described elsewhere (Krasnov *et al*. 2001, 2002a, 2003; Sarfati *et al*. 2005). An individual rodent host was placed in a glass cage (60 × 50 cm and 40 cm high) that contained a steel nestbox with a wire mesh floor and a pan containing a mixture of sand and dried bovine blood (larvae nutrient medium) on the bottom. We infested a host with 4–6 or 10–15 newly emerged *X. ramesis* (*D. dasyurus* or *M. crassus*, respectively) or six to eight newly emerged *P. chephrenis* (*A. cahirinus*) (initially with field-collected fleas). The gravid female fleas left the host and deposited eggs in the substrate and bedding material in the nestbox. After 2 weeks, we collected all substrate and bedding material from the nestbox and transferred it to an incubator (FOC225E, Velp Scientifica srl, Milan, Italy), where fleas developed at an air temperature of 25 °C and relative humidity of 75%. In this study, we used only newly emerged fleas, 2 days of age, which did not feed from emergence until experimental treatments.

Laboratory colonies of spiny mice and gerbils were established in 1999 from field-captured progenitors, which were collected from various locations in the Ramon erosion cirque, Negev Highlands, Israel (30°35′ N, 34°45′ E). Colonies were maintained in plastic cages (60 × 50 cm and 40 cm high) with sawdust bedding at 25 °C ambient temperature and 12D : 12L regime. Initially, each cage contained a male and a female. Young individuals were moved to a new cage at 2-months of age to prevent overpopulation and inbreeding. Newly formed groups also consisted of males and females from different parents. Millet seeds and alfalfa (*Medicago* sp.) were offered *ad libitum*. No water was available to the rodents as the alfalfa supplied enough water for their needs. Spiny mice were also offered commercial cat chow once a week. All rodents used in the experiments were adult (6–8 months old) males and were not initially infested by fleas. During experiments, these animals were maintained individually.

EXPERIMENTAL DESIGN

We randomly assigned 180 rodents of each species to three experimental groups, namely (a) pristine (140 individuals; never parasitized by fleas); (b) previously infested with *P. cheprenis* (20 individuals); and (c) previously infested with *X. ramesis* (20 individuals). Rodents from (b) and (c) groups were exposed to 50 fleas five times during 10 days prior to experiments (once in 2 days) (see details on exposure procedure below).

We randomly assigned 1400 newly emerged females and 600 newly emerged males of each species to the four experimental treatments that differed in (i) host species (a spiny mouse or a gerbil); and (ii) experience with flea parasitism (pristine or previously infested). In particular, *P. chephrenis* fed on spiny mice and gerbils from experimental groups (a) and (b), whereas *X. ramesis* fed on spiny mice and gerbils from experimental groups (a) and (c). Each treatment with each flea species was replicated 10 times.

EXPERIMENTAL PROCEDURES

An individual rodent was placed in a wire mesh (5 × 5 mm) tube (15 cm in length and 5 cm in diameter for spiny mice and 10 cm in length and 2 cm in diameter for gerbils) to prevent movement and self-grooming. Each tube with a rodent was placed in an individual white plastic pan. Thirty newly emerged female *X. ramesis* or *P. chephrenis* were weighed (± 0·01 mg, 290 SCS Precisa Balance, Precisa Instruments AG, Dietikon, Switzerland) and then released into the fur of a rodent, together with 15 conspecific males. Fleas were allowed to feed on rodents for 2 (*X. ramesis*) or 8 (*P. chephrenis*) hours (time of satiation estimated from our preliminary observations), and then were brushed out over a white plastic pan with a toothbrush. The hair of the rodent was brushed several times, until all fleas were recovered. The female fleas were then weighed as described above and the increase in body mass after parasitizing the rodents was considered to equal the amount of blood consumed by the fleas. After feeding, female and male fleas were placed in Petri dishes with a bottom covered by a thin layer of clean sand, transferred into an incubator (FOC225E, Velp Scientifica srl) and maintained at 25 °C air temperature and 90% RH. This procedure was repeated daily for 7 days. Each consecutive feeding of a flea group was done on a different randomly selected individual host of the same species. Each pristine rodent was used in experiments only once, whereas each previously infested rodent was used two to three times.

After each feeding, female fleas were examined under a light microscope and the degree of egg development was determined visually (see Krasnov *et al.* 2002b for details). Eggs started to develop on the third day of the study. Consequently, after the third feeding, small pieces of filter paper were added to Petri dishes with female fleas. Dishes and the filter paper were checked once a day during four consecutive days. We counted all eggs laid by all females in a given group during 4 days, recorded latency of oviposition (the first day of oviposition after start of the experiment) and calculated the mean number of eggs produced per female.

DATA ANALYSIS

Blood meal size was evaluated by calculating the amount of blood consumed per milligram body mass of fasting flea. The effects of host species and experience with fleas on blood meal size during 7 days of feeding were analysed using repeated measures ANOVAS with day of feeding as within-subject factor. The effects of these independent variables on egg production and latency of oviposition were analysed using two-way ANOVAS. All analyses were carried out separately for each flea species. Scheffe test was used for multiple comparisons. To adjust for deviations from normality, we applied a log-transformation on the dependent variables prior to analyses. Untransformed data are presented in figures.

Results

The size of a blood meal of *P. chephrenis* did not depend on either host species or previous infestation, but depended on the day of feeding (Table 1). The blood meal was significantly greater during the two last feedings than earlier feedings (*post hoc* comparisons for the effect of day factor, Scheffe tests, $P < 0.001$). However, an interaction between host species and day of feeding was marginally significant (Table 1). The effect of day of feeding on the blood meal size was manifested mainly when a flea fed on a gerbil rather than on a spiny mouse (*post hoc* comparisons for the effect of day × host species interaction, Scheffe tests, $P < 0.001$ and $P > 0.2$, respectively; Fig. 1a).

In contrast, the blood meal size in *X. ramesis* was affected by all independent variables (Table 1). In addition, all interactions between factors (except for the interaction between the factors of day and previous infestation) were significant. The manifestation of main effects was as follows. Fleas took the largest blood meal during the first feeding (*post hoc* comparisons for the effect of day factor, Scheffe tests, $P < 0.001$, except for the difference between the first and the last feedings; Fig. 1b). Furthermore, the blood meals consumed from a gerbil were larger than those from a spiny mouse (*post hoc* comparisons for the effect of host identity factor, Scheffe tests, $P < 0.001$). Fleas consumed significantly more blood from pristine than from previously infested gerbils (*post hoc* comparisons for the effect of host identity × previous infestation interaction, Scheffe tests, $P < 0.001$; Fig. 1b), but the amounts of blood taken from pristine and previously infested spiny mice were similar (*post hoc* comparisons for the effect of host species × previous infestation interaction, Scheffe tests, $P > 0.2$; Fig. 1b).

Host identity strongly affected egg production in both *P. chephrenis* and *X. ramesis* (Table 2). Both fleas produced

Table 1. Summary of repeated measures ANOVAS of the effect of host species (H; *Acomys cahirinus* vs. *Dipodillus dasyurus*), previous exposure of a host with fleas (E; pristine vs. previously infested) and day of feeding (D) on the amount of blood consumed per unit body mass of fasting flea

	Parapulex chephrenis				*Xenopsylla ramesis*			
Effect	SS	d.f.	*F*	*P*	SS	d.f.	*F*	*P*
H	0·001	1	0·30	0·59	0·080	1	60·26	< 0·001
E	0·001	1	0·07	0·79	0·013	1	9·48	< 0·01
H × E	0·001	1	1·21	0·28	0·007	1	5·29	0·03
Error	0·0211	36	–	–	0·055	36	–	–
D	0·0511	6	10·11	< 0·001	0·227	6	21·51	< 0·001
D × H	0·011	6	2·13	0·05	0·045	6	4·30	< 0·001
D × E	0·007	6	1·37	0·23	0·006	6	0·55	0·77
D × H × E	0·010	6	1·95	0·07	0·032	6	3·03	< 0·01
Error	0·182	216	–	–	0·433	216	–	–

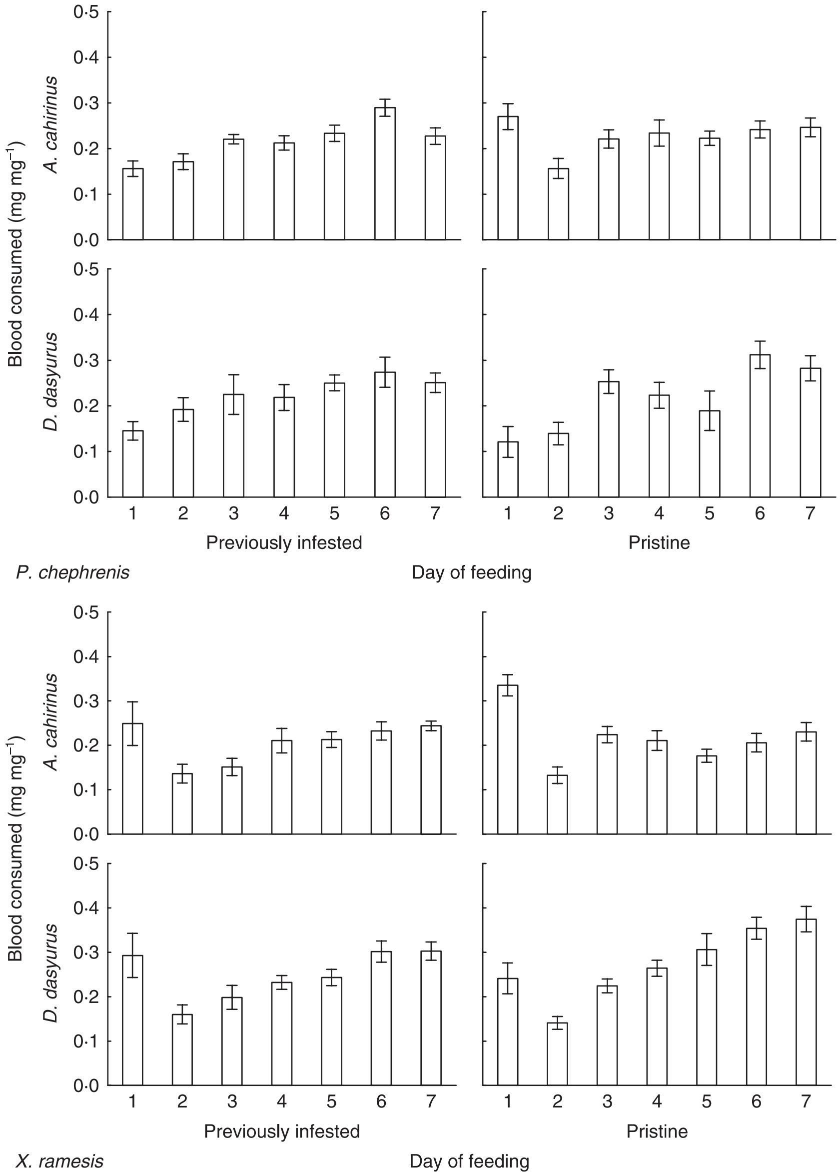

Fig. 1. Mean (± SE) amount of blood (mg per mg body mass of fasting flea) consumed by *P. chephrenis* and *X. ramesis* when feeding for 8 or 2 h, respectively, on pristine or previously infested *A. cahirinus* or *D. dasyurus*.

more eggs when they fed on their typical host, namely egg production of *P. chephrenis* was significantly higher when it fed on a spiny mouse (*post hoc* comparisons for the effect of host identity factor, Scheffe tests, $P < 0{\cdot}001$; Fig. 2a), whereas that of *X. ramesis* was higher when it fed on a gerbil (*post hoc* comparisons for the effect of host species factor, Scheffe tests, $P < 0{\cdot}001$; Fig. 3a). Latency of oviposition demonstrated the same trend with oviposition in *P. chephrenis* and *X. ramesis* starting earlier when they fed on spiny mice and gerbils, respectively (*post hoc* comparisons for the effect of host identity factor, Scheffe tests, $P < 0{\cdot}001$; Figs 2b and 3b), except in *P. chephrenis* feeding on pristine hosts (see below). An effect of previous infestation of a host with fleas on both egg production and latency of oviposition occurred in *P. chephrenis*, but not in *X. ramesis* (Table 2). Surprisingly, *P. chephrenis* laid more eggs and started oviposition earlier when it fed on previously infested rodents (*post hoc* comparisons for the effect of host identity factor, Scheffe tests, $P < 0{\cdot}001$; Fig. 2). In contrast, egg production in *X. ramesis* was similar and it started oviposition at approximately the same time, independent of whether rodents were infested with fleas prior to experiments (*post hoc* comparisons for the effect of host identity factor, Scheffe tests, $P > 0{\cdot}4$; Fig. 3). In addition, in *P. chephrenis*, interaction between independent variables was significant. The reason for this was that oviposition started earlier on previously infested spiny mice than on previously infested gerbils, but no difference in latency of oviposition between pristine hosts of different species was found (*post hoc* comparisons for the effect of host identity × previous infestation interaction, Scheffe tests, $P < 0{\cdot}001$ and $P > 0{\cdot}6$, respectively).

Table 2. Summary of ANOVAS of the effect of host species (H; *Acomys cahirinus* vs. *Dipodillus dasyurus*) and previous exposure of a host with fleas (E; pristine vs. previously infested) on the number of eggs produced by a female flea and days to oviposition

	Parapulex chephrenis				*Xenopsylla ramesis*			
Effect	SS	d.f.	*F*	*P*	SS	d.f.	*F*	*P*
Egg production								
H	0·894	1	94·15	< 0·001	1·577	1	164·031	< 0·001
E	0·330	1	38·35	< 0·001	0·015415	1	1·60	0·21
H × E	0·015	1	1·73	0·20	0·001	1	0·01	0·91
Error	0·309	36	–	–	0·394	36	–	–
Latency of oviposition								
H	0·0226	1	17·83	< 0·001	0·031	1	7·57	< 0·01
E	0·0226	1	17·83	< 0·001	0·001	1	0·08	0·78
H × E	0·008	1	6·26	0·02	0·001	1	0·01	0·97
Error	0·046	36	–	–	0·157	36	–	–

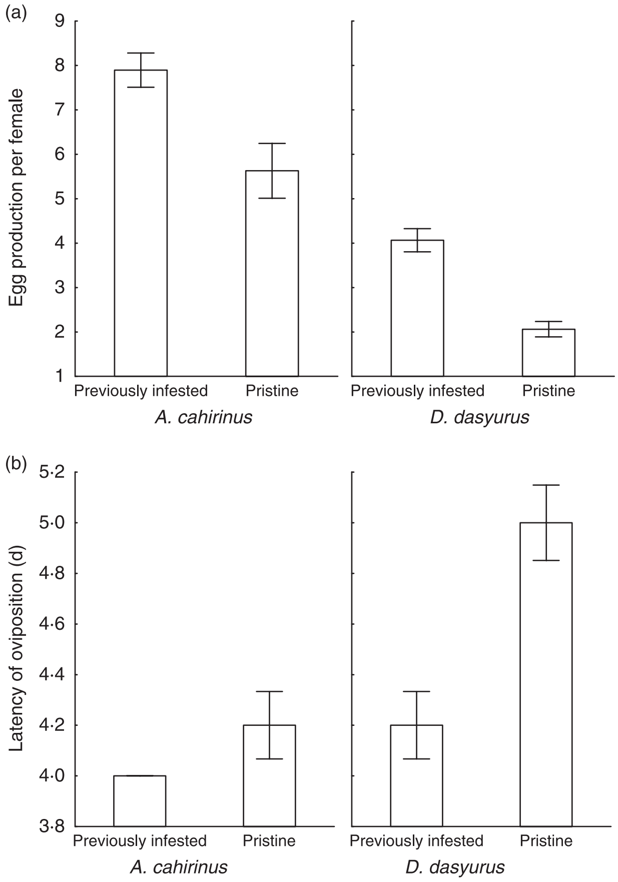

Fig. 2. Mean (± SE) (a) number of eggs produced per female and (b) latency of oviposition (days from start of the feeding to the first egg) of *P. chephrenis* feeding on pristine or previously infested *A. cahirinus* or *G. dasyurus*. Missing SE whiskers denote invariant results.

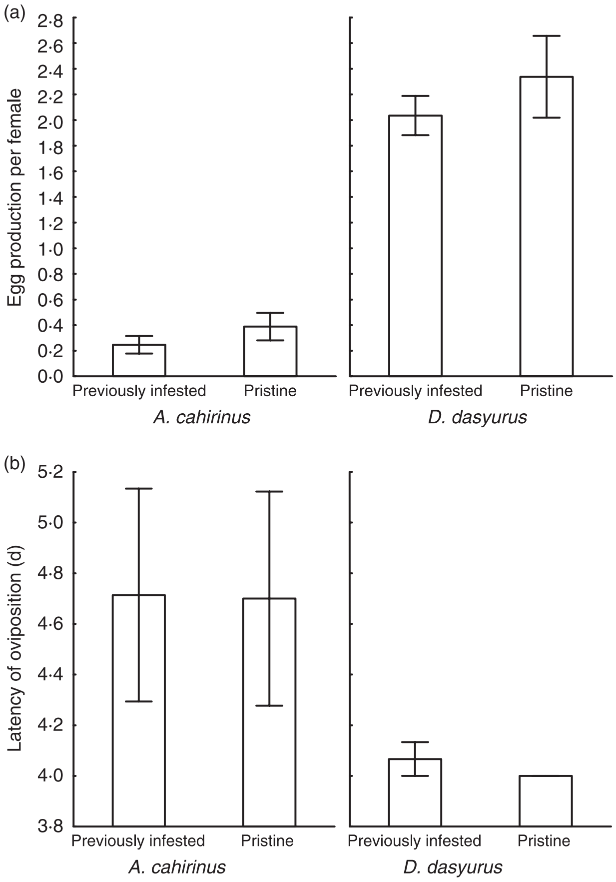

Fig. 3. Mean (± SE) (a) number of eggs produced per female and (b) latency of oviposition (days from start of the feeding to the first egg) of *X. ramesis* feeding on pristine or previously infested *A. cahirinus* or *G. dasyurus*. Missing SE whiskers denote invariant results.

Discussion

This study supported some of our predictions. Higher feeding success in a typical host was found in *X. ramesis*, but not in *P. chephrenis*, whereas both fleas responded to host species, as predicted, in terms of their egg production and latency of oviposition. The only evidence for acquired resistance after repeated attacks by fleas was a higher feeding efficiency of *X. ramesis* on pristine than on previously infested gerbils. Surprisingly, the reproductive pattern of *P. chephrenis* on pristine and previously infested hosts of either host species was in the opposite direction to that predicted. Finally, between-flea comparison of the effect of host species and its previous exposure to fleas on their feeding and reproduction patterns did not support the prediction on the low sensitivity of the host's immune system to attacks of a specific parasite.

EFFECT OF HOST SPECIES

The physical and chemical properties of the host's blood as well as host skin structure are important characteristics to which a host-specific flea is adapted (Marshall 1981; Lehane 2005). It is not surprising, therefore, that feeding of a flea is strongly affected by the identity of the host species (Seal & Bhattacharji 1961; Prasad 1969; Kamala Bai & Prasad 1979; Liu *et al*. 1993). Although we did not find between-host difference in blood meal size of *P. chephrenis* in this study, the effect of host species on feeding performance of this flea was found when other feeding variables such as the rate of satiation and digestion were measured (Krasnov *et al*. 2003; Sarfati *et al*. 2005). Perhaps, measures such as blood meal size during a prolonged period of feeding are not sensitive enough to indicate subtle differences in responses to host species in some flea species.

Nonetheless, *X. ramesis* demonstrated a between-host difference in feeding performance as expected. This suggests that even a host opportunist performs differently on different host species in terms of feeding (see also Krasnov *et al*. 2004 for *X. conformis* and Bibikova & Gerasimova 1967 for *X. skrjabini*).

Interestingly, the temporal pattern of feeding differed between the two flea species. *Parapulex chephrenis* consumed more blood at late feeding bouts than at early feeding bouts, whereas *X. ramesis* consumed the largest amount of blood at the first feeding. This difference likely reflected differences in some life-history traits between *P. chephrenis* and *X. ramesis*, as well as those between their preferable hosts. *Parapulex chephrenis* stays on a host for prolonged periods, whereas the contact of *X. ramesis* with a host is usually intermittent and only for a blood meal (N. V. Budelova & B. R. Krasnov, unpublished data). In addition, the Egyptian spiny mice (typical host of *P. chephrenis*) tend to nest communally with most individuals residing in their home ranges for a long period (Shargal, Kronfeld-Schor & Dayan 2000), whereas Wagner's gerbils are solitary and/or abandon their burrow for long time periods and change it often (Gromov, Krasnov & Shenbrot 2000). Consequently, it would be advantageous for *X. ramesis* to consume as much blood as possible at the earliest feeding occasion because its host may not always return in a regular or predictable manner to its burrow. In contrast, a source of food for *P. chephrenis* is almost always guaranteed. The increase in blood meal size at the end of the 7-day feeding period could be compensation for energy expenditure for egg maturation.

A strong effect of the host species on the reproductive success of both host-specific and host-opportunistic fleas supports our previous results (Krasnov *et al*. 2002a, 2004), that flea preference of a particular host species is reflected in an increase of fitness when feeding on that host compared to other hosts. Similar findings were reported in other studies (e.g. Seal & Bhattacharji 1961; Prasad 1969; Gerasimova 1973). It should be noted, however, that egg production is only one component of fitness. Testing patterns of survival and development of offspring produced by fleas feeding on different hosts may provide clearer results about fitness consequences of host preferences. For example, although the dog fleas, *Ctenocephalides canis*, laid eggs after feeding on cats, their offspring did not develop beyond the first larval stages (Baker & Elharam 1992).

EFFECT OF PREVIOUS INFESTATION WITH FLEAS

The most surprising results of this study were that *P. chephrenis* produced more eggs and started oviposition earlier when feeding on previously infested than on pristine hosts. These findings were opposite to those expected if exposure of hosts to flea parasitism induces acquired resistance against fleas. Furthermore, no effect of previous infestation with fleas on feeding and reproductive performance was found in *X. ramesis*. There can be several, not necessarily alternative, reasons for these findings. First, it is possible that hosts developed acquired resistance but it was not efficient. Indeed, cellular changes (e.g. basophilic responses) in the skin of guinea pigs and rats had no effect on feeding success of *X. cheopis* (Vaughan, Jerse & Azad 1989). The long history of adaptations of a haematophagous parasite to immune responses of a host could result in the development of some mechanisms that allow the parasite to cope successfully with the immune defence of a host. Inadequate efficiency of acquired immunity coupled with the high energetic cost of immune responses (e.g. Sheldon & Verhulst 1996) can be a reason why energy-limited hosts or hosts subjected to intense and/or diverse parasite attacks often give up their immune defence and surrender to parasites (Jokela, Schmid-Hempel & Rigby 2000; Krasnov *et al*. 2005b). Second, at least for *P. chephrenis*, it is possible that innate rather than acquired host's responses might have a negative effect on flea reproduction, whereas further flea exploitation leads to suppression of the immune function and results in an increase of flea reproductive performance (e.g. Khokhlova *et al*. 2004). Third, as mentioned above, egg production and latency of oviposition are only components of fecundity but cannot be considered as explicit measures of lifetime fecundity. For example, it is possible that previous exposure of a host to parasites may affect parasite fitness negatively in terms of quality (e.g. viability and development rate) but not quantity of offspring.

At first glance, the effect of the host's previous infestation with fleas in *P. chephrenis* feeding on both rodents did not support the results of earlier studies that demonstrated different immune responses of hosts to natural and unnatural parasite species (e.g. Vaughan *et al*. 1989). However, these studies measured host variables whereas, in our study, flea variables were measured. Perhaps, there is an asymmetry in host vs. parasite responses when a host responds to a parasite attack, whereas a parasite either does not respond to host defences or is able to overcome them. Nevertheless, it was reported that typical hosts of some tick species did not acquire immunity (measured via tick variables) against these ticks (e.g. Rechav & Fielden 1997). This was probably due to the presence of various substances in the tick saliva which allowed successful evasion of host immune responses (Ribeiro 1987) and which fleas (at least *P. chephrenis*) may lack.

Finally, in contrast to *P. chephrenis*, egg production and latency of oviposition of *X. ramesis* were similar, independent of whether the host was previously parasitized. This between-flea difference may reflect, to some extent, the difference in their host specificity which, essentially, is the difference in the level of ecological specialization. As a result, any difference between hosts, that is, not only in their species affinity but also in the degree of their experience to parasites, might cause sharper responses in a host-specific than in host-opportunistic parasite.

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